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RESEARCH ARTICLE

Joint Bias Correction and Downscaling of Subseasonal Forecasts via Diffusion Models

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Abstract

Subseasonal forecasts provide valuable guidance for decision-making, but remain limited by systematic biases and coarse spatial resolution, particularly over regions with complex orography. Quantile mapping (QM) is widely used for post-processing because it enables effective bias correction and downscaling, yet it applies a distributional correction and therefore neglects the case-specific dependence between forecasts and observations. To move toward a more integrated approach, we explore a diffusion-based generative framework for simultaneous probabilistic bias correction and high-resolution downscaling, with lead time explicitly included as a conditioning variable. We employ a conditional denoising diffusion model trained on high-resolution gridded observations (2 km) and conditioned on coarse-resolution ensemble-mean subseasonal forecasts (36 km) from the ECMWF IFS subseasonal system. We focus here on a univariate temperature model and on training strategies. By learning conditional distributions of observations given forecasts and lead time, diffusion-based generative models enable case-dependent corrections that exploit forecast–observation relationships. Two training configurations are examined to assess how lead-time conditioning and data availability influence the learning of biases. Results show that the diffusion model recovers realistic fine-scale temperature patterns and achieves meaningful lead-time-dependent bias correction across subseasonal lead times, even when trained on relatively limited datasets. However, compared to QM, the diffusion-based forecasts exhibit lower skill in probabilistic metrics such as the Continuous Ranked Probability Score and discrimination for extremes. Overall, this work represents a first step toward lead-time-aware subseasonal post-processing using diffusion models and highlights directions for future methodological development.

Impact Statement

Subseasonal forecasts play an increasingly important role in decision making for sectors such as energy planning, agriculture, and public health. However, these forecasts are typically produced at coarse spatial resolution and exhibit systematic biases, limiting their direct applicability for impact-oriented uses, particularly in regions with complex terrain. This study introduces a diffusion-based generative approach that performs lead-time-aware spatial downscaling and probabilistic bias correction of subseasonal temperature forecasts within a unified framework. Unlike conventional post-processing methods such as quantile mapping, which correct forecasts based on marginal distributions and ignore case-specific forecast–observation relationships, the proposed method learns a conditional distribution of high-resolution temperature fields given the coarse-resolution forecast and lead time, enabling case-dependent probabilistic post-processing. This enables

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a computationally efficient and scalable pathway for operational implementation. The framework is demonstrated over Switzerland, where strong elevation gradients and local climate effects pose substantial challenges for forecasting systems. While traditional statistical methods remain competitive in probabilistic skill, this work establishes diffusion-based generative modeling as a viable foundation for next-generation subseasonal post-processing. By identifying current capabilities and remaining limitations, the study outlines clear directions for future development toward high-resolution subseasonal predictions that can better support decision making.

1. Introduction

Subseasonal forecasting, typically defined as the prediction of atmospheric and surface variables from 2 weeks to 2 months ahead, fills a critical gap between medium-range weather prediction (up to two weeks ahead) and seasonal forecasts. This timescale is of high socioeconomic relevance for water resources management, wildfire and drought mitigation, agriculture, and disaster preparedness (Pappenberger et al., 2015; Pulwarty and Sivakumar, 2014; White et al., 2022). Despite notable progress over the past decade (Pegion et al., 2019; Domeisen et al., 2022; Vitart et al., 2025), subseasonal forecasting remains a challenge (Merryfield et al., 2020).

Operational numerical weather prediction (NWP) systems have extended their forecast horizons into the subseasonal range using large global ensembles, typically operating at horizontal resolutions of 30–50 km (e.g., ECMWF IFS, NASA GEOS-S2S). This resolution reflects a compromise between computational constraints and the need to represent large-scale coupled processes, such as the Madden–Julian Oscillation (MJO), which are key sources of subseasonal predictability (Vitart, 2017). However, systematic model biases in circulation, teleconnections, and surface variables grow with lead time and substantially degrade forecast skill across the subseasonal range (Pegion et al., 2019). In parallel, large-scale data-driven forecasting systems, including Pangu-Weather, GraphCast, and specialized subseasonal architectures such as FuXi-S2S, TianQuan-Climate, and AIFS-S2S, have demonstrated improved representation of precipitation and teleconnections at lead times up to 45 days (Bi et al., 2023; Chen et al., 2023; Wang et al., 2024; Lang et al., 2024). Nevertheless, like traditional NWP systems, these models remain constrained by relatively coarse spatial resolution.

Most subseasonal forecasting systems are designed to capture large-scale circulation patterns and, therefore, operate at resolutions insufficient to resolve mesoscale processes such as convection, diabatic heating, and flow–orography interactions, which occur at kilometer scales (Bauer et al., 2015). Errors arising from unresolved processes can propagate upscale, weakening large-scale teleconnections (e.g., MJO) and reducing forecast skill, particularly in the extratropics (Ham et al., 2019; Jia et al., 2019; Lin et al., 2022). These limitations are especially pronounced in regions of complex terrain, where sharp elevation gradients require kilometer-scale resolution for accurate representation (Lundquist et al., 2019). As a result, the lack of high-resolution, bias-corrected subseasonal forecasts remains a barrier for impact-oriented applications such as flood and drought prediction, renewable energy forecasting, and heat-stress assessment.

Deep generative models, and diffusion models (DMs) in particular, have emerged as powerful tools for probabilistic climate downscaling and bias correction (Vandal et al., 2022; Mardani et al., 2024). Conditional DMs learn the distribution of fine-scale variability given coarse-scale forcing, enabling the generation of physically plausible high-resolution fields while explicitly representing uncertainty. Unlike deterministic post-processing approaches, DMs can implicitly correct systematic

biases (Hatanaka et al., 2023) and preserve multivariate dependencies when trained jointly (Lopez-Gomez et al., 2025), avoiding inconsistencies inherent in traditional sequential bias-correction and downscaling pipelines (Maraun, 2013; Ehret et al., 2012).

Motivated by these advances, we develop a diffusion-based generative framework that simultaneously performs spatial downscaling and lead-time–dependent bias correction of ECMWF IFS subseasonal 2-meter temperature forecasts while generating a 50-member ensemble to represent forecast uncertainty. We evaluate the approach over Switzerland, a region characterized by pronounced orographic complexity spanning the Swiss Plateau and the Alpine mountain range. Strong elevation gradients and land–atmosphere interactions provide a stringent test for high-resolution subseasonal forecasting, particularly during summer, when convective processes dominate uncertainty. This setting enables a rigorous assessment of the ability of diffusion-based models to recover fine-scale structure and uncertainty from coarse, biased subseasonal forecasts.

2. Data and methods

2.1. Data and Preprocessing

Observational data

The high-resolution target field used to train the diffusion model (DM) is the 2 km gridded daily mean 2 meter temperature (2t) over Switzerland, derived from the MeteoSwiss TabsD dataset (MeteoSwiss, 2021). This observational product is generated through the spatial interpolation of approximately 90 homogenized long-term meteorological stations. The interpolation methodology incorporates a non-linear estimation of vertical temperature profiles and the use of non-Euclidean distances, enabling a more accurate representation of temperature patterns influenced by cold-air pooling and downslope winds in mountainous valleys. The TabsD gridded record begins in 1961 and is continuously updated. At the time of data acquisition, data were available until 2020.

ECMWF subseasonal hindcasts

The coarse-resolution conditioning input for the diffusion model consists of ECMWF subseasonal hindcasts. Subseasonal hindcasts are retrospective weather forecasts (re-forecasts) generated for past dates using the same numerical model and configuration as current real-time operations. ECMWF produces hindcasts to establish a robust *model climate*, which serves as a baseline for identifying significant anomalies, especially after the first prediction days. Hindcasts are used rather than real-time forecasts in order to maximize the size and consistency of the training dataset, which is essential for learning systematic model biases. To align with the availability of the observational dataset, we use ECMWF subseasonal hindcasts covering the period 2002–2021 from IFS cycle 47r3 (operational in 2022). This hindcast (and forecast) version is initialized twice weekly (Mondays and Thursdays) and extends up to 46 forecast days, with a native horizontal resolution of approximately 18 km for days 1–15, 36 km for days 16–46, and 137 vertical levels (see ECMWF Implementation of IFS Cycle 47r3 for details). The publicly available subseasonal hindcasts were distributed by ECMWF for download at the horizontal resolution of 36 km. Since cycle 47r3, several substantial upgrades have been implemented in the IFS, including the assimilation of 2-meter temperature observations, changes in the data assimilation systems, improvements to the land surface model, and a doubling of the number of ensemble members (Ingleby et al., 2024), leading to improvements in forecast skill (Roberts et al., 2024). Despite these advancements in the IFS, subseasonal forecasts still face persistent biases and resolution constraints, especially in topographically diverse regions. This study provides a framework designed to learn and correct these errors dynamically. As such, the approach remains highly relevant for current and future operational cycles.

Quantile mapped subseasonal forecasts

The benchmark used for comparison in this study is the statistical downscaling framework employed

operationally by MeteoSwiss for subseasonal forecasting. This framework utilizes Empirical Quantile Mapping (QM) to bias-correct and downscale ECMWF subseasonal 2t ensemble forecasts against the TabsD high-resolution observational field, following the approach validated by Monhart et al. (2018). Specifically, the technique maps the distribution of the raw model output onto a 2 km climate observational grid to produce localized forecasts.

Forecast verification metrics

To evaluate forecast performance we employ a suite of standard meteorological verification metrics as detailed in (Wilks, 2011). The forecast Mean Error (ME) provides a primary assessment of deterministic accuracy and systematic error, indicating whether the ensemble mean consistently over- or under-predicts the observed values. To assess the quality of the probabilistic ensemble, we use the Continuous Ranked Probability Score (CRPS), which generalizes the Mean Absolute Error (MAE) to probability distributions by integrating the squared difference between the forecast and observed cumulative distributions. Finally, the Receiver Operating Characteristic (ROC) Area Under the Curve (AUC) score is used to measure the system's discrimination ability, in our case specifically, its capacity to distinguish between the occurrence and non-occurrence of temperature extremes above the observed 90th percentile. To further support the results of this work, additional verification diagnostics are provided in the Supplementary Material due to manuscript length constraints. These include the Energy Score (ES), which extends the CRPS to multivariate settings and evaluates the overall quality of ensemble forecasts, and rank histograms, which assess ensemble reliability by indicating whether the observations are statistically consistent with the forecast distribution.

2.2. Methodology

Generative Diffusion Framework

We employ a high-resolution meteorological downscaling model based on the Conditional Denoising Diffusion Probabilistic Model (DDPM) framework (Mardani et al., 2024). The model is designed to generate $\mathbf{y}$, a high-resolution 2-meter temperature field, conditioned on $\mathbf{x}$, a low-resolution ensemble mean input from the ECMWF subseasonal system, by learning the full conditional probability distribution $p(\mathbf{y} \mid \mathbf{x})$. During the diffusion process, structured noise is introduced, which, unlike standard independent Gaussian noise, accounts for the spatial correlations and multi-scale dependencies inherent in atmospheric fields. By iteratively removing this noise during inference, the model generates physically plausible fine-scale details, such as those driven by local topography, that are statistically consistent with high-resolution observations.

Network Architecture and Training

The core of our system is a U-Net backbone enhanced with the "Elucidating the Design Space of Diffusion-Based Generative Models" (EDM) framework (Karras et al., 2022). We use EDM preconditioning to ensure training stability across varying noise scales. The input to the network is a two-channel tensor consisting of the noisy target image and the low-resolution condition $\mathbf{x}$, upscaled via bilinear interpolation and concatenated along the channel dimension. By using bilinearly upscaled forecasts as the conditioning input, we provide the DM model with a spatially smooth but physically plausible starting point. In this configuration, the DM is explicitly tasked with a dual objective: first, to correct the systematic magnitude errors (bias) inherent in the low-resolution model data, and second, to recover the high-frequency spatial details (downscaling) absent in both the original low-resolution model and the subsequent interpolation step. The current verification evaluates the combined impact of these tasks, as the model is specifically architected to address both simultaneously. Additionally, we use lead time as a scalar auxiliary conditioning label to allow the model to learn lead-time-dependent bias patterns. Rather than predicting absolute temperature fields, the network is trained to model the residual between the bilinearly upscaled coarse input and the high-resolution reference. For numerical stability and faster convergence, this residual is standardized using a global Z-score normalization,

where the mean and variance are computed across all grid points and lead times in the training set. The error scale is homogenized while preserving the underlying lead-time-dependent bias patterns and spatial structures. This allows the model to leverage the auxiliary lead-time conditioning to learn and correct systematic errors without being hindered by the high absolute magnitude of the raw temperature values. The model is trained using a noise-weighted Mean Squared Error (MSE) loss, which prioritizes the denoising performance at noise scales most critical for atmospheric feature reconstruction.

Probabilistic Sampling and Inference
To generate the final bias-corrected and downscaled product, we use a second-order deterministic sampler (Heun's method) with 40 steps. This allows us to transform Gaussian noise into high-resolution temperature fields that respect the large-scale constraints of the NWP input while adding realistic mesoscale variability. Starting from different random noise seeds for the same NWP input $\mathbf{x}$, we generate a 50-member ensemble. This ensemble is used to quantify the forecast uncertainty and provide probabilistic bias correction, effectively mapping the biased, coarse-grained dynamical model output to the observational data.

Training stategy
The ECMWF hindcast dataset comprises 520 hindcasts, each consisting of 11 ensemble members (one control and ten perturbed) initialized over 20 summers (June, July, August) from 2002 to 2021. We adopt a stratified training approach, in which the DM is trained separately for each model start date (i.e., initialization date). This strategy is chosen to account for the strong subseasonal non-stationarity of the atmospheric state; by isolating start dates, the model can more effectively learn the specific seasonal evolution of biases and land-atmosphere feedbacks unique to each phase of the summer. While a common alternative would be to pool all start dates and utilize a seasonal conditioning label, we chose separate training for each start date to maximize the model's sensitivity to date-specific atmospheric regimes. This strategy avoids the risk where the model might average the distinct physical biases of early-summer snow-melt periods with late-summer dry regimes (Orth and Seneviratne, 2015). For each start-date, we train for 3000 epochs using the ensemble mean 2t fields from 2002–2017 without incorporating additional climate variables. Model selection and hyperparameter tuning are performed using the 2018–2019 validation period, while the year 2020 is reserved as an independent test set. Moreover, note that even though the target consists of daily 2t fields, the evaluation on the test period is aggregated to a weekly scale in order to enable a consistent assessment of subseasonal forecast skill.

Motivated by the lead-time dependent biases reported for subseasonal forecasts over Switzerland (i.e., Figure 12 in Monhart et al. (2018)), we evaluate two distinct training strategies. The first, an all-lead-time approach, utilizes the ensemble mean fields across the full 46-day forecast horizon, incorporating daily lead-time information as a conditioning variable during training. The second, a week-1-restricted approach, excludes lead-time information and limits the training data strictly to lead-week 1. To compensate for the significantly reduced sample size in the week-1-restricted approach, we increase the training duration to 6,000 epochs to ensure model convergence. For the all-lead-time approach we train for 24 hours using 4 GPUs, while for the week-1-restricted approach we train for 48 hours. The training was performed on Nvidia Grace Hopper processors in MeteoSwiss' Research and Development cluster Balfrin, at CSCS (Swiss National Supercomputing Centre).

3. Results and Discussion

Both training strategies demonstrate a similar capability to downscale the ECMWF forecasts, producing spatially detailed temperature fields that exhibit patterns similar to those observed in the reference dataset (Fig. 1). In terms of bias correction (Fig. 2), complete bias removal remains challenging for all methods at these forecast ranges, particularly given the limited training data and the inherent uncertainty at longer lead times. Both DM approaches provide notable improvements of mean error (ME)

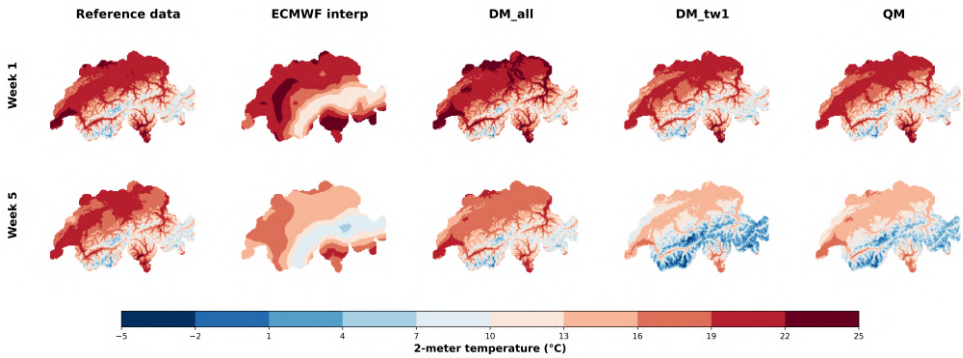

Figure 1. *Weekly 2t from example forecasts initialized on 15 August 2020 (columns 2-5), together with the reference observational dataset (column 1). The panels display forecasts from the interpolated ECMWF IFS cycle 47r3 (second column), the DM_all approach (third column), the DM_tw1 approach (fourth column), and the QM approach (fifth column). The results are given for lead week 1 (first row) and lead week 5 (second row).*

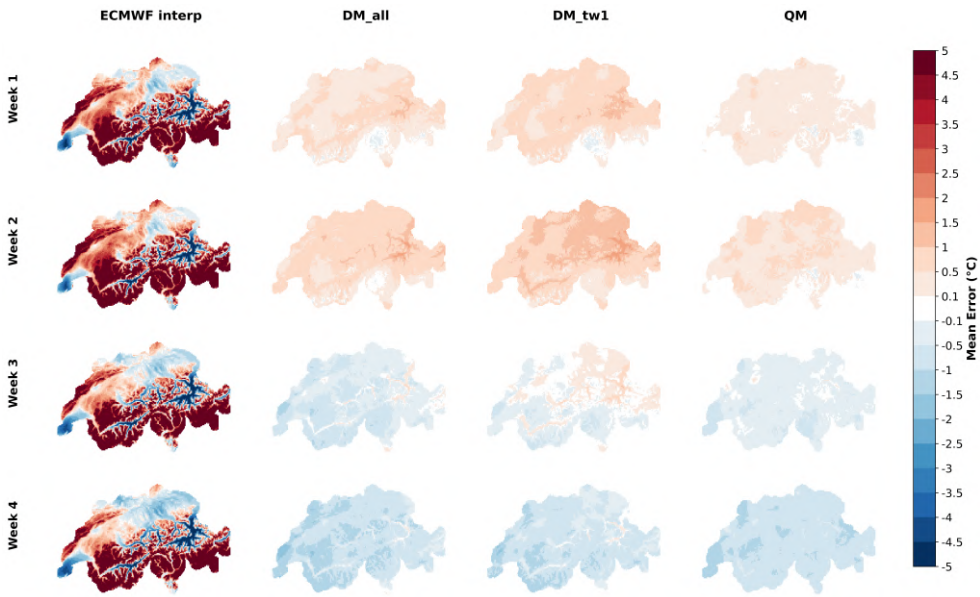

Figure 2. *Weekly 2t ME derived from all model initializations between June-August 2020 for the interpolated ECMWF IFS cycle 47r3 (first column), the DM_all approach (second column), the DM_tw1 approach (third column), and the QM approach (fourth column). ME values closer to zero indicate a better score.*

over the interpolated ECMWF data, although the improvements are somewhat smaller, though still comparable, to those achieved by QM. For lead weeks 1 and 2, the all-lead-time strategy (DM_all) outperforms the week-1-restricted approach (DM_tw1) over most of Switzerland, as indicated by smaller values of ME over most areas (Fig. 2). In particular, for week 1 predictions, the all-lead-time approach reduces the ME by roughly twice as much as the week-1-restricted strategy, suggesting that the diffusion model effectively exploits information from the daily week-1 lead times. For weeks 3 and 4, as well as for longer lead times not shown here (weeks 5 and 6), the week-1-restricted approach achieves

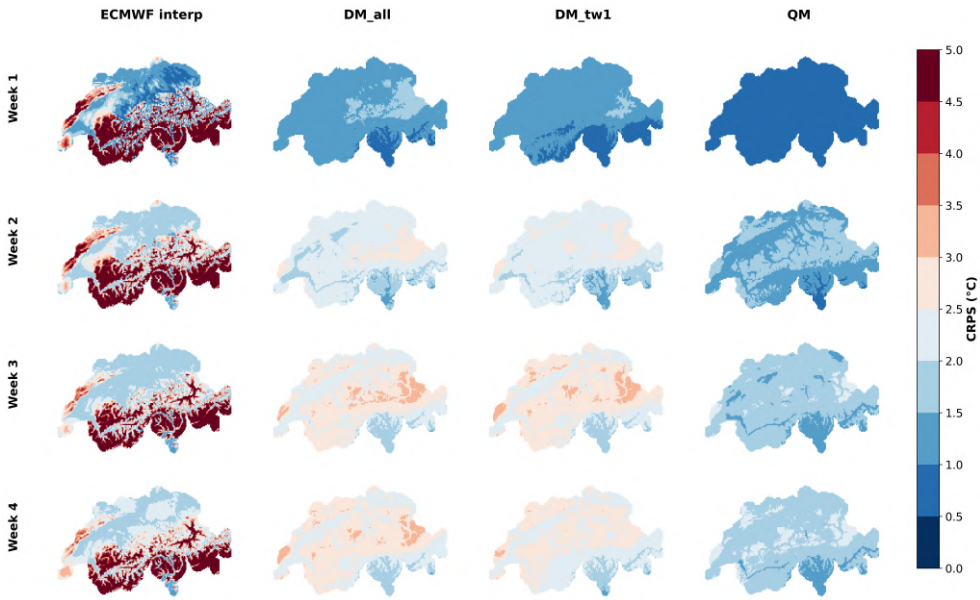

Figure 3. *As Figure 2, but for the weekly 2t CRPS. CRPS values closer to zero indicate a better score.*

a bias correction that is comparable to or slightly better than that of the all-lead-time strategy. However, as the differences between the two DM approaches are small for longer lead times a larger sample size would be required to robustly assess their relative bias-correction performance. The similar bias correction observed for longer lead times between the two training strategies is partly attributable to the relative stability of weekly mean biases in the interpolated ECMWF forecasts at lead times longer than 2 weeks and to the small bias differences across the different lead weeks for several Swiss locations (see supplementary Figure S1). Specifically, the bias can remain relatively stable either across all lead times (e.g., Zurich and Geneva in Figure S1) or between shorter (first two weeks) and longer lead times (e.g., Locarno and Sion in Figure S1). Although the DM_tw1 approach is trained on a smaller dataset than DM_all, it benefits from longer training time and is still exposed, through week-1 data, to bias magnitudes comparable to those at longer lead times. Moreover, the diffusion model learns the overall data distribution rather than performing simple location-to-location adjustments, which likely contributes to its ability to generalize across lead times. It is also important to emphasize that the inputs to the diffusion model consist of ECMWF forecasts interpolated from 36 km to 2 km resolution. This interpolation step can amplify the raw bias at all lead times, particularly over orographically complex regions. Moreover, the ECMWF model outputs are compared against observational data that were not seen during training. Consequently, discrepancies arising from resolution differences and the use of independent observational references may have a larger impact on model errors than the lead-time-dependent drift itself. The observed bias stability at specific locations and/or lead times is consistent with the results of Dutra et al. (2021) (see their Figure 3a,b), that evaluate subseasonal ECMWF forecasts against ERA5, the dataset used for the ECMWF model initialization (Hersbach et al., 2020; Vitart et al., 2019).

The CRPS of the raw ECMWF forecasts is substantially reduced by both DM approaches, particularly over orographically complex regions, although the reduction is generally smaller than that achieved by QM (Fig. 3). For lead week 1, the QM method reduces the raw ECMWF CRPS by nearly 1 °C over the Swiss Plateau and by more than 5 °C in complex terrain. The DM approaches reduce the ECMWF CRPS by approximately 0.5 °C over the Swiss Plateau and by about 4 °C over orographically complex regions, where the raw ECMWF CRPS reaches about 1 °C and exceeds 5 °C, for these

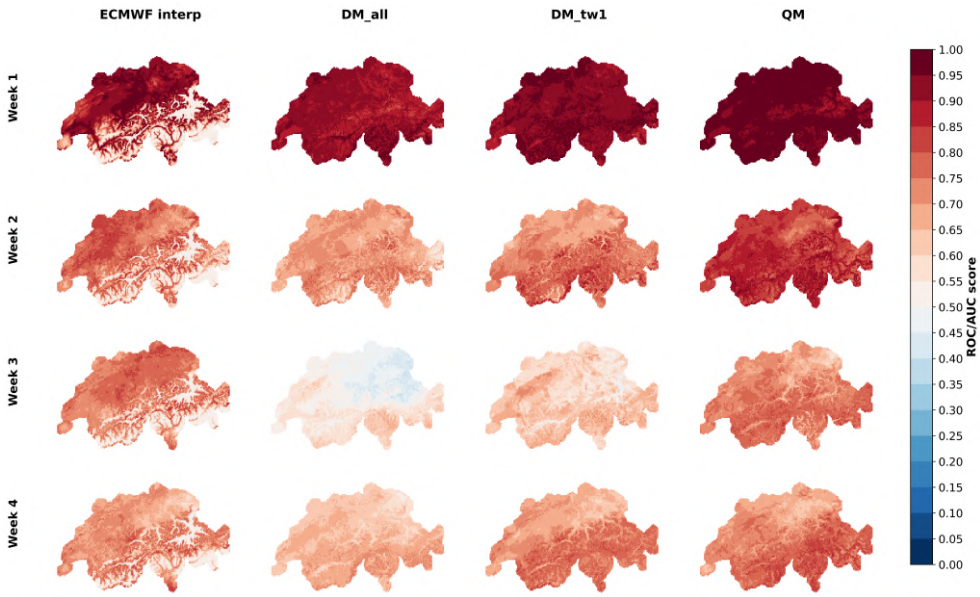

Figure 4. *As Figure 2, but for the weekly ROC/AUC score of 2t extremes above the observed 90th percentile. ROC/AUC values closer to 1 indicate a better score and values equal or lower to 0.5 indicate no skill beyond random chance in discriminating between extreme and non-extreme events.*

regions respectively. While the all-lead-time DM approach yields marginally lower CRPS improvements compared to the week-1-restricted strategy, the magnitude of the differences is small to support a statistically robust distinction between the two methods.

Finally, we assess the ability of the different approaches to predict hot extremes, defined as temperatures exceeding the observed 90th percentile (Fig. 4). Discrimination skill is evaluated using the ROC/AUC score, where values of 0.5 or lower indicate no skill beyond random chance. The ROC/AUC metric accounts for both the hit rate and the false-alarm rate across all possible decision thresholds, making it well suited for evaluating probabilistic predictions of extremes. Across all lead weeks, the QM approach attains AUC scores exceeding 0.7 in most of Switzerland. Although slightly lower, both DM strategies also demonstrate skill, with regional AUC values consistently above 0.5 for all lead weeks and both approaches, except for the all–lead-time strategy at lead week 3. From lead weeks 1 to 4, the week-1-restricted DM strategy generally outperforms the all–lead-time approach, whereas for lead weeks 5 and 6 (not shown), the all–lead-time strategy exhibits better discrimination. Across all methods, the ROC score generally decreases with increasing lead time from 1 to 6 weeks; however, both DM strategies display an unusually large reduction at lead week 3. A similar anomaly at lead week 3 is also evident in the CRPS (Figure 3) and may be related to the change in forecast resolution at this lead time in the ECMWF system (see Methods). For lead week 1, QM achieves the highest AUC, exceeding 0.95 for nearly all Swiss regions, followed by the DM approaches with regional values mostly above 0.9. This indicates that while QM performs best overall, both DM methods retain substantial skill in discriminating hot extremes across lead times.

Although DM_all is exposed to a larger training set, the inclusion of longer lead times (weeks 2–6), which are characterized by increased ensemble spread, results in a smoother ensemble mean used as training input. The competitive performance of DM_tw1 for longer lead times suggests that, despite being trained on a smaller dataset, it effectively leverages the higher signal-to-noise ratio of week-1 data. This indicates that the diffusion model is able to learn the fundamental relationship between large-scale atmospheric conditions and fine-scale topographic features primarily from these high-quality

inputs. Importantly, this learned relationship appears to generalize well to later lead times, even though the model is not explicitly trained on them.

4. Conclusion

This study introduces a generative diffusion-based framework for the simultaneous downscaling and bias correction of subseasonal temperature forecasts over complex terrain. By conditioning a probabilistic diffusion model on coarse-resolution ECMWF subseasonal forecasts and explicitly incorporating lead-time information, we bias correct and downscale coarse horizontal resolution (36 km) ECMWF subseasonal hindcasts into high horizontal resolution (2 km) daily mean temperature fields over Switzerland. The proposed approach generates physically plausible, high-resolution temperature fields while simultaneously accounting for forecast uncertainty through ensemble sampling. Both of these aspects are important for impact-oriented applications.

We have evaluated two distinct training strategies on a univariate 2t model. Specifically, the all-lead-time approach reduces the mean error in week-1 predictions by approximately 50% more compared to the week-1-restricted strategy. For longer lead times, both methods show comparable performance, likely due to the relative stability of weekly mean biases in the underlying coarse resolution forecasts as compared to the reference dataset that was never seen by the model and is based on spatially interpolated observations. While the baseline Quantile Mapping (QM) method retains an advantage in terms of CRPS and extreme event discrimination (ROC/AUC), the diffusion models successfully reduce raw ECMWF CRPS by up to 4 °C in orographically complex regions. QM's strong performance reflects its targeted optimization of marginal distributions, making it a difficult baseline to outperform, particularly in probabilistic verification metrics.

A key contribution of this work is the explicit examination of lead-time conditioning and training strategies in generative downscaling for the subseasonal range. By contrasting an all-lead-time training strategy with a week-1-restricted approach, we show that diffusion models can flexibly correct bias structures across lead times, while remaining robust even when trained on relatively limited datasets. Importantly, our results indicate that effective subseasonal post-processing does not necessarily require extensive training data across all lead times, making the approach attractive for applications where data availability or computational resources are constrained. Beyond forecast skill, the proposed framework offers advantages for operational use. Unlike traditional quantile mapping, the diffusion model generates its own ensemble from a single forecast input, reducing storage needs and enabling relatively fast inference (50 members in 40 minutes on a single GPU). In addition, its generative design allows new predictors or variables to be easily incorporated, facilitating extension to other surface variables or challenging regimes such as wintertime temperature inversions.

While this study focuses on summer temperature forecasts over Switzerland, chosen as a challenging test case due to strong orographic effects and pronounced summer biases, the methodological insights are broadly transferable. The proposed diffusion-based framework is not tied to a specific model cycle, geographic region, or atmospheric variable, and is therefore expected to remain applicable as dynamical forecasting systems continue to evolve. In this work, the diffusion model is conditioned on the ensemble-mean forecast in order to isolate and systematically evaluate the impact of the two training strategies on bias correction performance. This design choice, however, may limit the generalization of the model and its ability to fully represent ensemble spread.

Several promising avenues for future development, therefore, emerge from this study. Training the diffusion model on all ensemble members or on a single forecast (e.g., control forecast) could improve the representation of forecast uncertainty and extremes and potentially enhance probabilistic skill. In addition, jointly training across multiple initialization dates may improve generalization across atmospheric regimes while retaining lead-time awareness. Further extensions could include multivariate downscaling, incorporation of additional physical predictors, or the prediction of impact-relevant indices. Although these developments are beyond the scope of the present work, they represent

natural and well-motivated next steps toward fully exploiting diffusion-based generative models for subseasonal forecasting.

Overall, this work provides a concrete and operationally relevant benchmark for applying lead-time informed diffusion-based generative models to subseasonal forecasting. It demonstrates that such models can effectively bridge the gap between coarse-resolution dynamical forecasts and the high-resolution probabilistic information required for impact-oriented applications, thereby contributing a practical tool to the growing field of climate informatics.

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Competing Interests. The authors confirm that they have no competing interests.

Data and Code Availability Statement. ECMWF IFS hindcast data are available from the ECMWF public data archive: (https://apps.ecmwf.int/archive-catalogue/) The observational TabsD dataset can be accessed through Meteoswiss' Open Government Data infrastructure (https://www.meteoschweiz.admin.ch/service-und-publikationen/service/open-data.html) The DM code is publicly available at: (https://github.com/marpyr/Diffusion_Downscaling)

Ethical Standards. The research meets all ethical guidelines, including adherence to the legal requirements of the study country.

Author Contributions. M.P. conceptualized and conducted the study, built the DM model, analyzed and visualized the data, and wrote the original manuscript draft. A.I. provided the hindcast data and quantile mapped forecasts, and conceptualized the study. D.B. conceptualized the study. C.S. and D.D. acquired funding, conceptualized, and supervised the study. All authors reviewed the manuscript.

Supplementary Material. Supplementary material is available in a separate document.

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